

Piter Z. Garcia Bautista

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Research Profile

Graduate researcher focused on machine learning for diagnostics and decision support under noisy, heterogeneous, and partially observed data. Background includes an M.S. in Computer Science, current M.S. work in Data Science, industry healthcare AI experience, and computational biology training. Current work emphasizes reproducibility, robust evaluation, fairness-aware analysis, and translational relevance to clinical settings.

Research Interests

Machine learning for precision medicine and diagnostics; robust learning under missing or uncertain context; fairness-aware and trustworthy AI for high-stakes decision support; reproducible computational evaluation systems.

Education

Rochester Institute of Technology

Master of Science, Data Science — GPA 3.9

Expected Aug 2026

Rochester, NY

- Relevant coursework: Bioinformatics Algorithms, Data-Driven Knowledge Discovery, Neural Networks, Applied Statistics, Software Engineering for Data Science.

Rochester Institute of Technology

Master of Science, Computer Science — GPA 3.2

2015

Rochester, NY

- Concentrations: Artificial Intelligence, Big Data Analytics, Computer Graphics.

University of Rochester, Warner School of Education

Graduate study, Teaching Computer Science K–12 (M.S. track) — GPA 4.0

In progress

Rochester, NY

- NSF Robert Noyce Scholar; Advanced Certificate in Leadership in Disability and Inclusive Practices.

Research and Professional Experience

Rochester Institute of Technology, Software Engineering Department

Graduate Assistant, Quantum and AI Research

Fall 2025 – Present

Rochester, NY

- Build reproducible machine learning evaluation pipelines across local, Colab, and GCP environments with structured logging and validation.
- Develop experimental bandit and decision-making systems for uncertain, heterogeneous settings, with emphasis on reliability and traceability.
- Produce manuscript-ready technical analyses and research artifacts for faculty-led projects.

Rochester Institute of Technology

2026

Computational Biology Research Projects (BIO614 and BIO550)

Rochester, NY

- Led substantial computational work in BIO614, including formal manuscript development in the project *BIO614-FinalProjectProposal* (Overleaf), emphasizing algorithm implementation, structured evaluation, and scientific reporting.
- Complemented BIO614 with BIO550 RNA-seq and differential-expression analysis, including reproducible preprocessing, quality-control validation, and biological interpretation.

VEDADATA Inc.

Sep 2022 – Aug 2024

Data Solutions Engineer

New York, NY

- Designed scalable Python and pandas workflows for ingestion, validation, preprocessing, and analytics operations.
- Strengthened production data reliability through structured quality checks and statistical diagnostics.

VIOME Inc.

Jan 2021 – Sep 2022

AI Data Solutions Engineer

New York, NY

- Developed healthcare-oriented machine learning workflows for preprocessing, feature engineering, model experimentation, and evaluation.
- Supported reproducible analytics workflows in AWS-based environments for predictive health applications.

Selected Technical Writing

- *Fairness-Aware Bandits for Network Routing in Quantum and Clinical Settings* (DSCI601 project proposal, manuscript-style report).
- *BIO614-FinalProjectProposal* (Overleaf) and companion BIO550 computational biology analyses on RNA structure and RNA-seq workflows, with reproducible diagnostics-oriented interpretation.
- *Equitable Bioinformatics: Enhancing Diagnostic Decision-Making through RNA and Biomarker Data* (ISTE-780 Project Report) — systematic fairness assessment of ML-driven diagnostic algorithms, with reproducible SHAP-based bias detection and statistical significance testing.

Technical Competencies

Programming: Python, R, Java, C++, C#, SQL, MATLAB, JavaScript, Bash

Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, model evaluation, uncertainty-aware analysis, fairness-aware ML

Data and Statistics: pandas, NumPy, SciPy, statistical modeling, reproducible benchmarking

Research Tooling: Git/GitHub, Linux, Docker, LaTeX, AWS, GCP

Selected Fit for KTH Single-Cell Position

- Combined computational biology evidence from BIO614 and BIO550, including manuscript-style technical writing and RNA-seq differential-expression analysis.
- Strong Python-based machine learning and reproducibility background for robust method development on high-dimensional, heterogeneous biomedical data.

- Clear research motivation for AI and machine learning methods that improve interpretability and reliability in single-cell cancer-data analysis.